Mini-project

CardioKey (ECGID)

Analog ECG Signal Acquisition & Biometrics

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Abstract

Traditional biometric methods (fingerprint, face ID) are vulnerable to spoofing and external modification. Passwords and tokens can be stolen or lost.

We propose to build a analog implimentation of Electrocardiogram (ECG) as a biometric identifier. The core of ECGID lies in the unique anatomical and physiological characteristics of an individual's heart, which dictate the specific morphology of the P, QRS, and T waves.

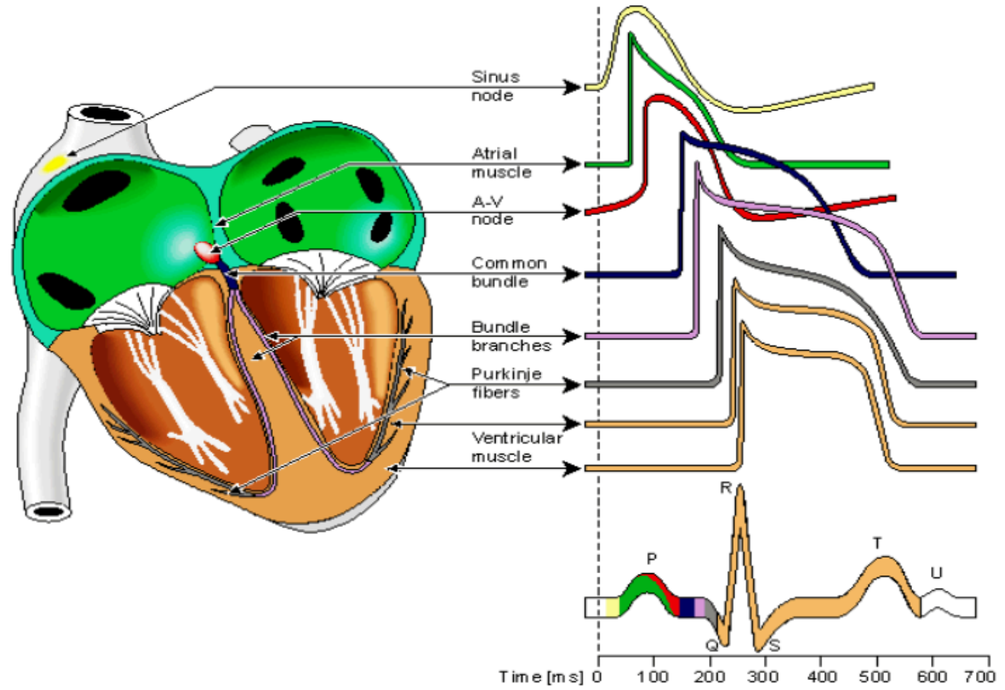


Fig. 1. the heart's anatomy with waveforms from different specific part of the heart [19].

1. ECG & Characteristics

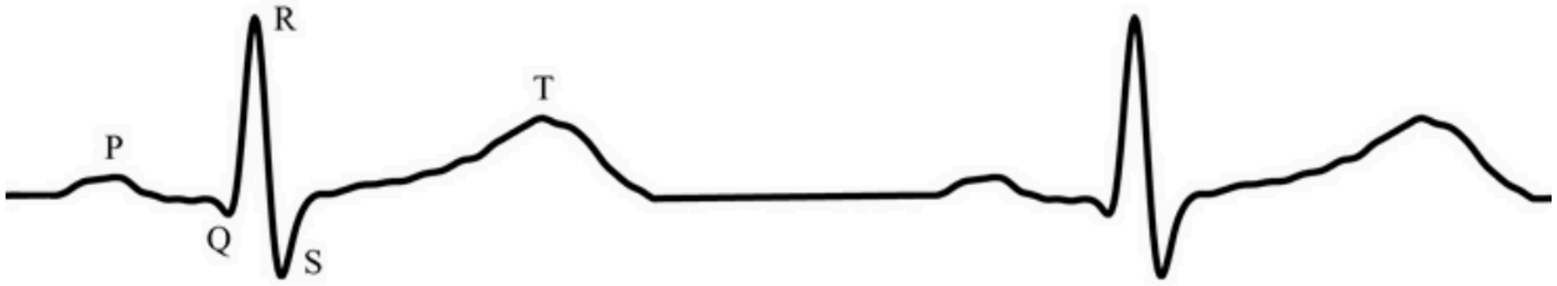


Figure 1. Example of ECG with standard notations.

two cardiac cycles

Muscle contraction causes depolarization, which is detected as voltage changes on the skin's surface over time.

P Wave

Represents the first upward deflection and atrial depolarization

QRS Complex

Composed of Q, R, and S waves; represents ventricular depolarization. This is the most distinctive fragment for identification.

T Wave

Represents ventricular repolarization, concurrent with the end of ventricular systole.

U Wave

A series of waves following the T wave, discernible in some tracings.

key points of biometric identification

1. Universality : Every living human generates ECG signals continuously.

2. Uniqueness: Waveforms are influenced by heart location, size, geometric shape, chest structure, age, and physiological characteristics.

3. Stability: Adult heartbeat structures are basically fixed; waveforms remain stable over time unless lesions or major pathologies (like myocardial infarction) occur.

4. Measurability: Modern equipment is miniaturized and portable, allowing for high-precision acquisition. eg : using 12 lead electrodes for accuracy

ECG variability due Heart rate varibility → shortening of diastole duration (the baseline fragment of the ECG) and the ventricular depolarization period (corresponding to the ST segment in the ECG) and attenuation of the R wave amplitude but QRS complex remains constant

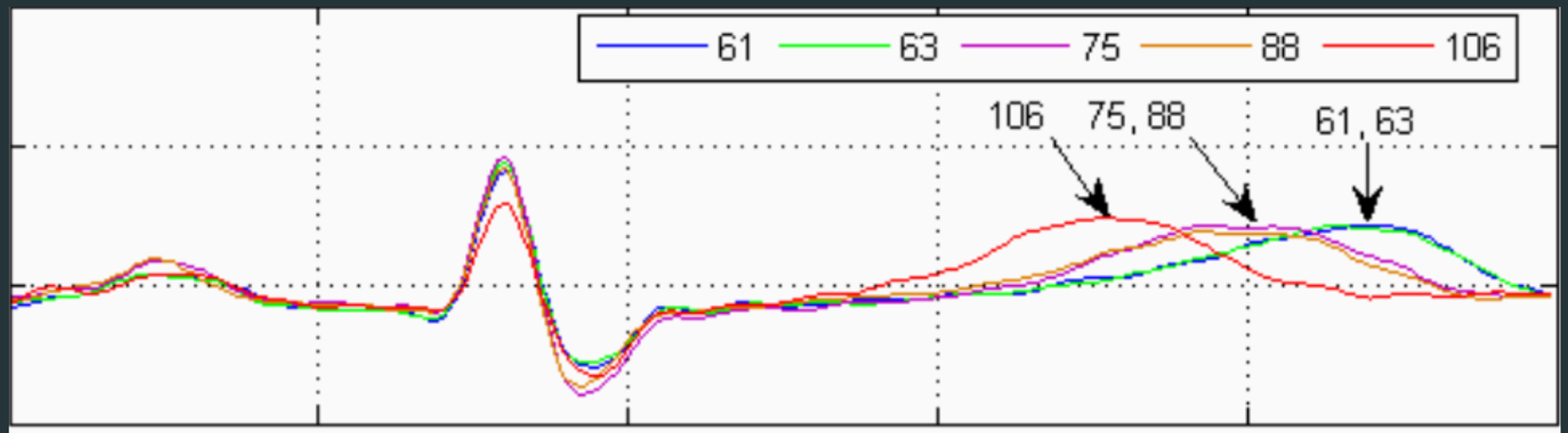
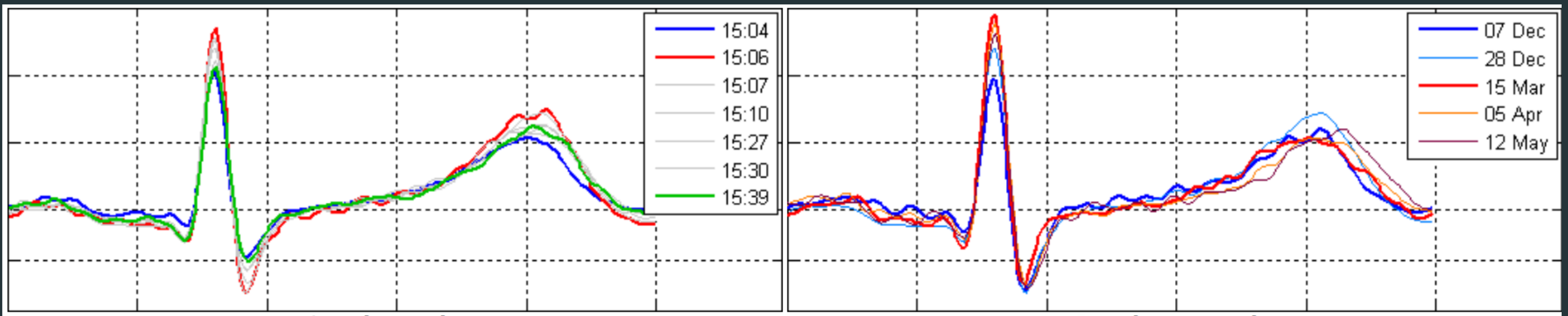


Figure 3. Variation of ECG of one individual with different heart rates.



A. within one hour

B. within six months

Figure 4. Variation of ECG of one individual recorded at different times.

there are many other natural and artificial, intentional and unintentional causes of ECG variability. For example, taking certain medications may temporarily change the configuration of the cardiac cycle

Methodology

The system utilizes a precision analog front-end based on the AD620 instrumentation amplifier to acquire low-amplitude heart signals. This signal is digitized and processed using digital signal processing (DSP) techniques to extract the P-QRS-T complex.

Hardware components : ECG Analog Signal Acquisition

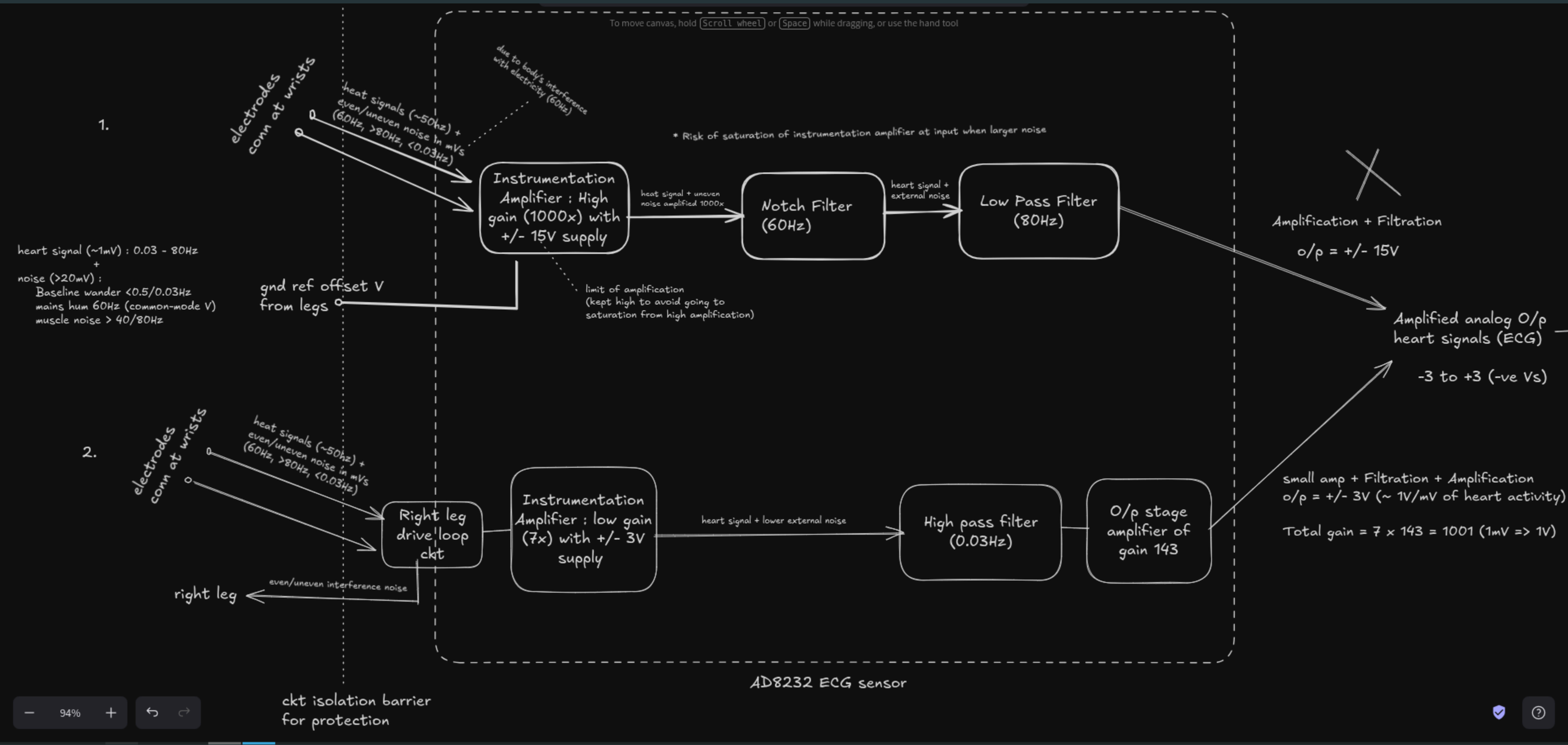
3 electrodes (1Lead LA-RA, feedback), AD620 instrumentation amp, AD705J, IC741 opamps for filters and amplifiers, and resistors & capacitors → AD8232 ECG sensor
Diodes for DC offset circuit, AD115 16bit ADC, a MCU (ESP32)

Software components : ECG Digital Signal Analysis & Biometrics

MATLAB or Python, TinyML and algorithms for signal analysis

2. Analog ECG Signal Acquisition - Hardware

two approaches



Right leg drive circuit

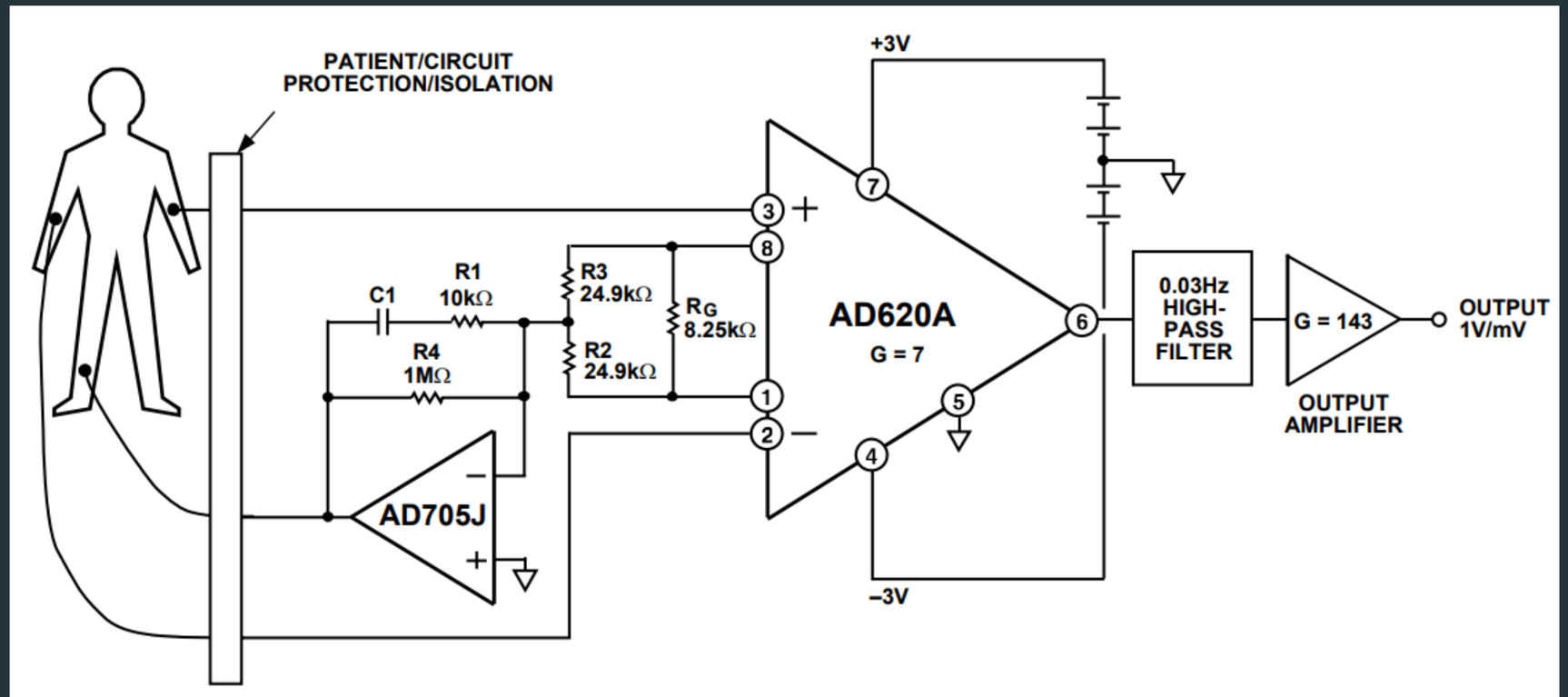


Figure 39. A Medical ECG Monitor Circuit

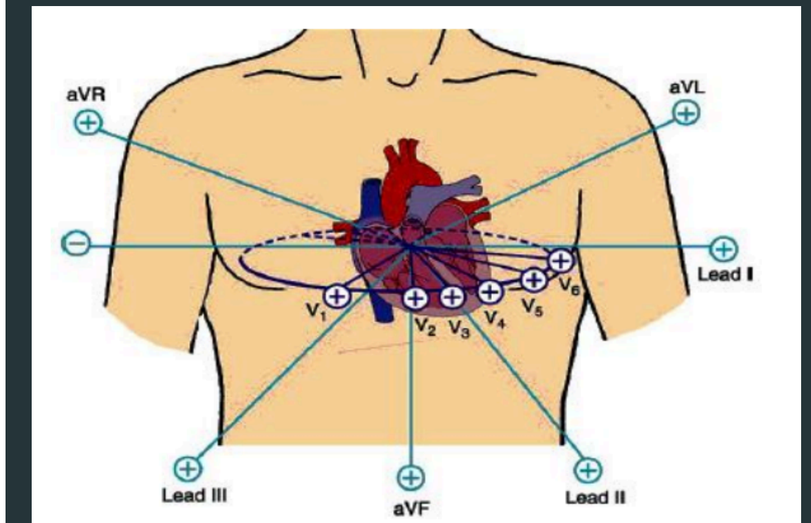


Fig. 3. A vector view of the standard 12 Lead ECG [23].

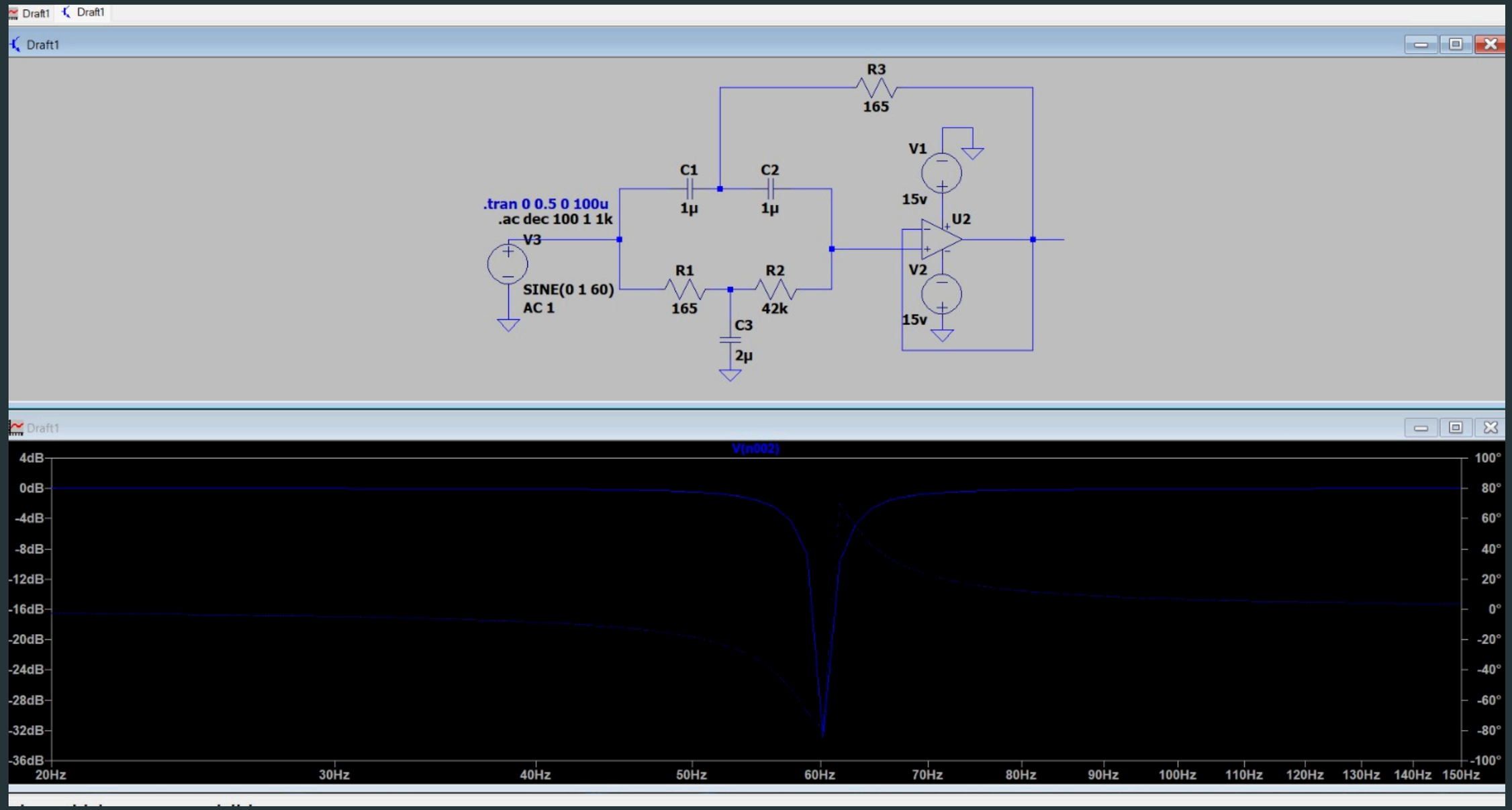
Right Leg Drive feedback circuit (60Hz mains hum & >80Hz muscle noise) → AD620 instrumentation amplifier (common mode rejection) → High pass filter (baseline wander) → output amplifier

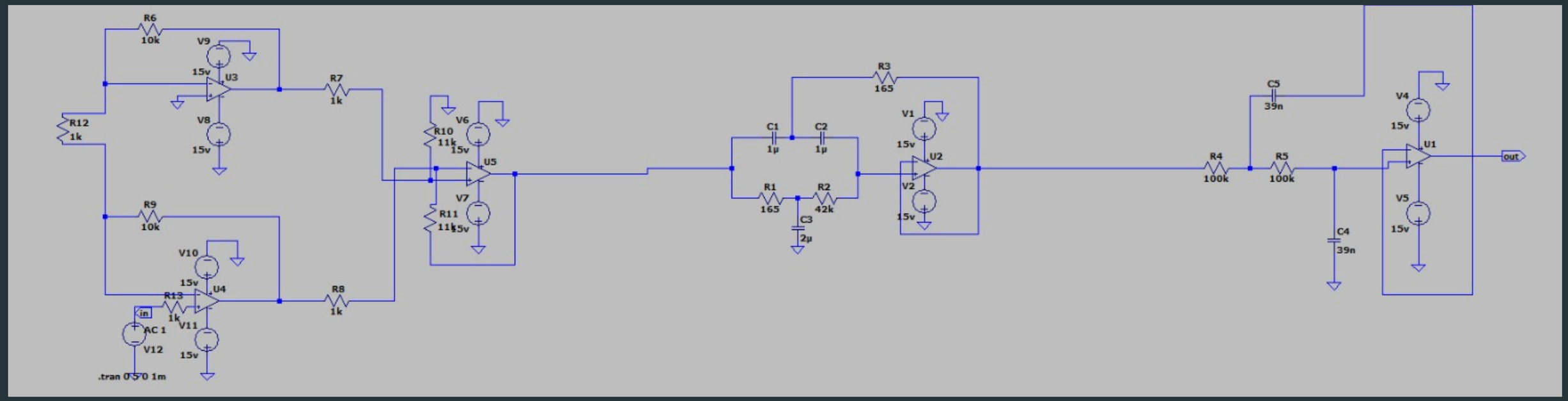
Total Gain = 7 x 143 = 1001 ie 1V/mV of heart signal (0.03 Hz - 80Hz)

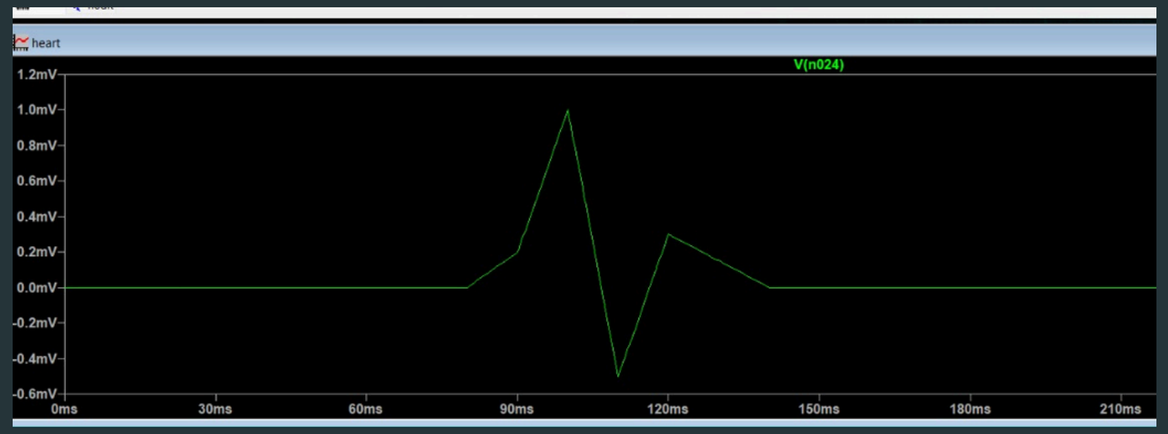
Vlead1 = LeftArm - RightArm → heart signal
Vavg = (-) (LA+RA)/2 → noise

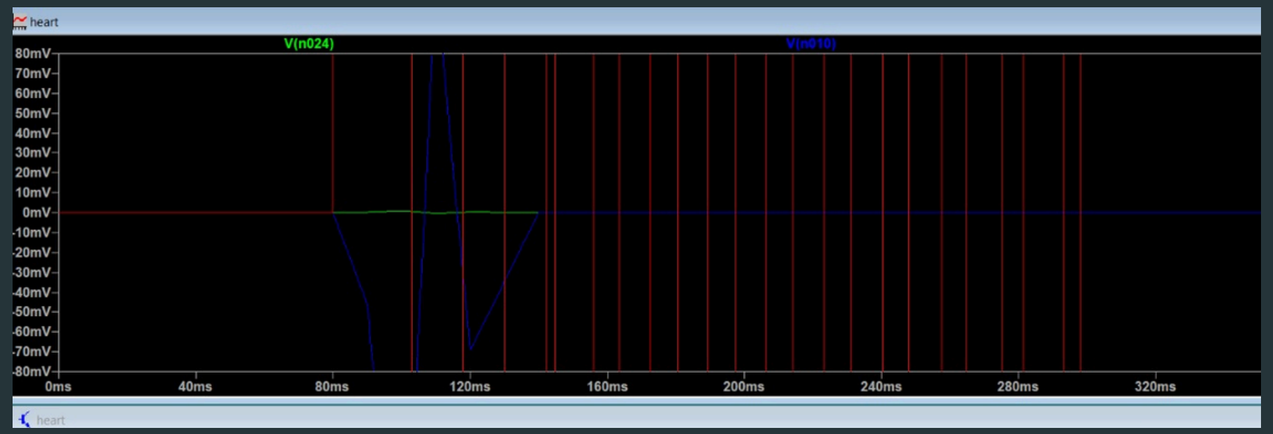
Simulation & Tests :

Notch filter attinuating 60 Hz

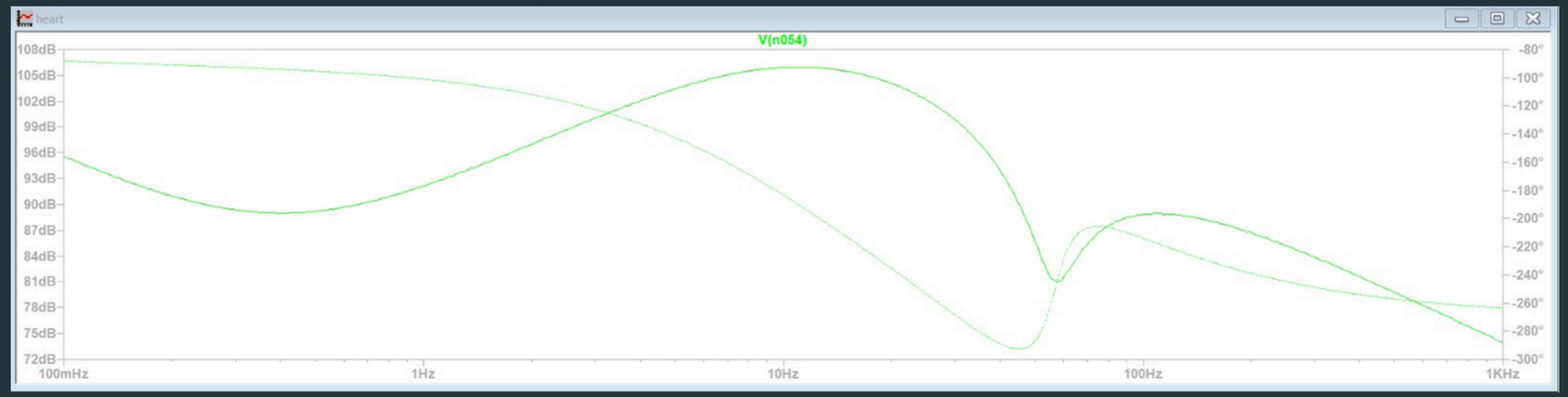




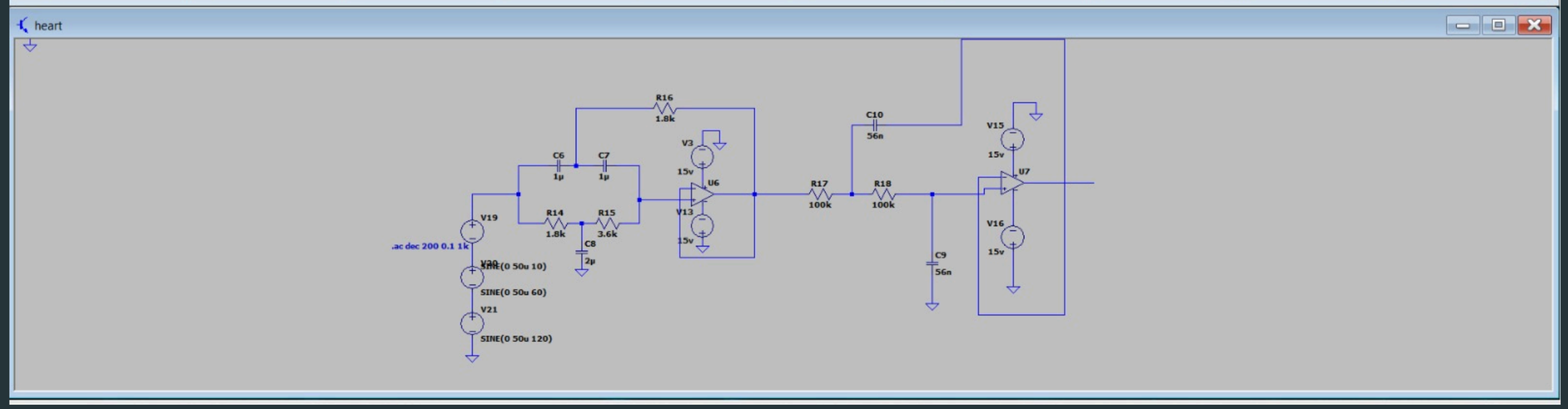




first approach (noise hitting saturation)



notch filter + low pass (transient analysis)



3. ECG Digital Signal Analysis - Software

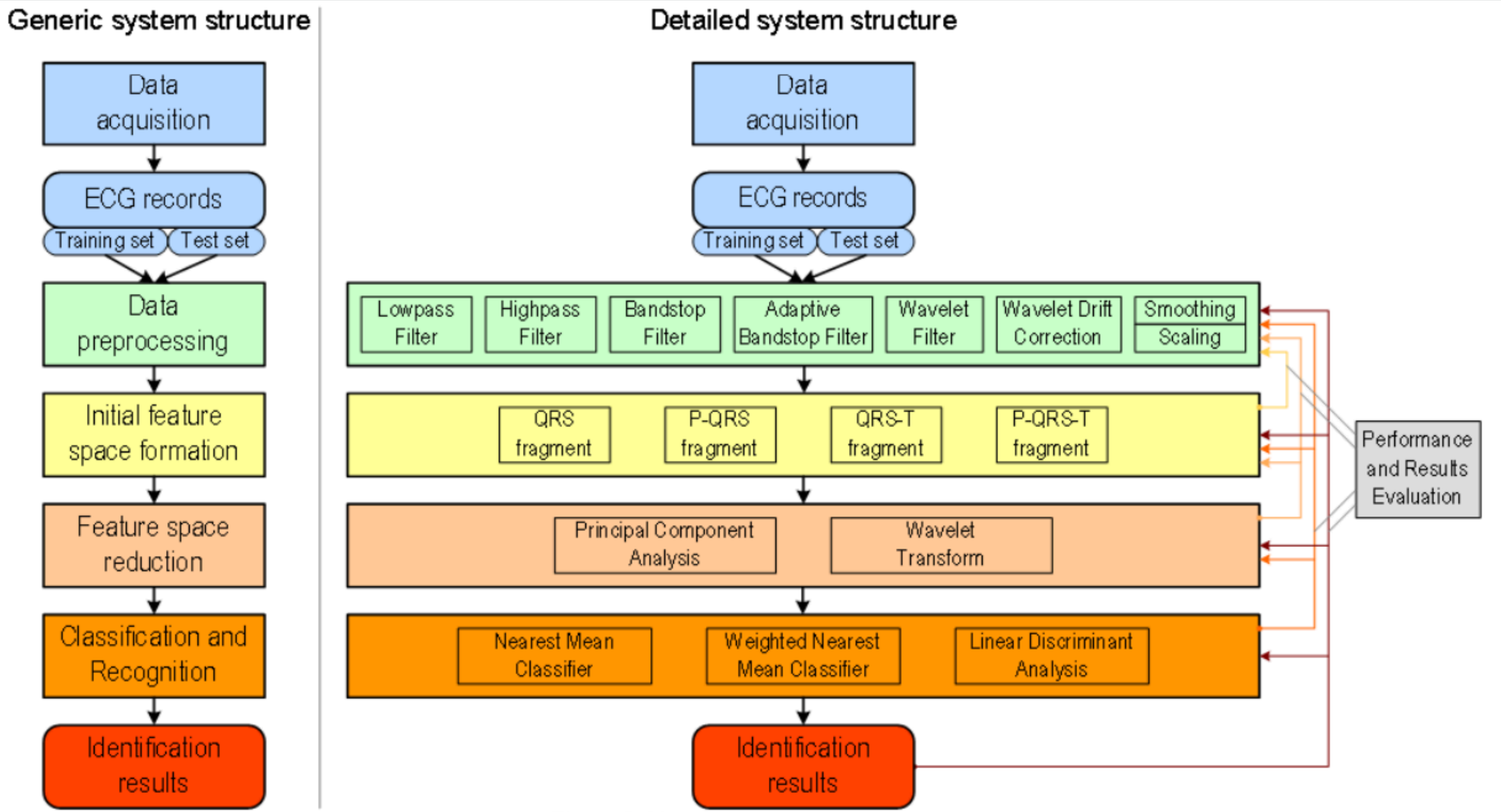
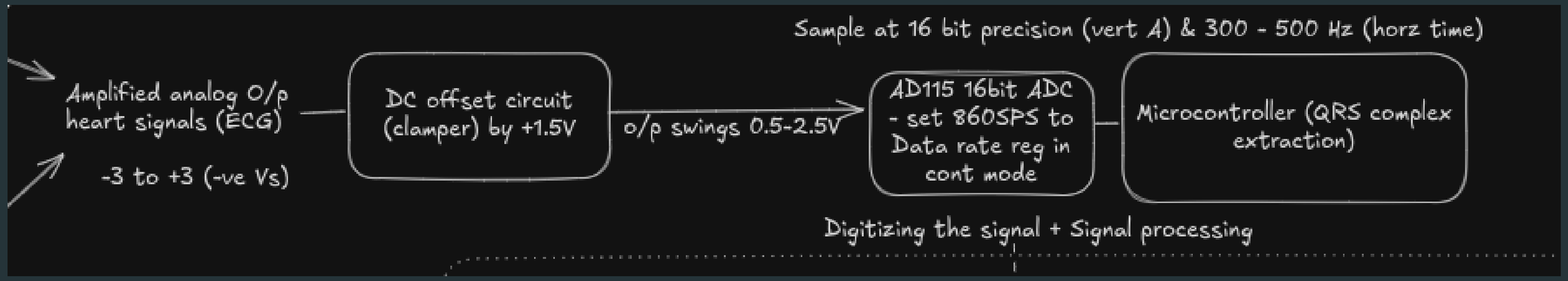


Figure 5. Identification system structure.



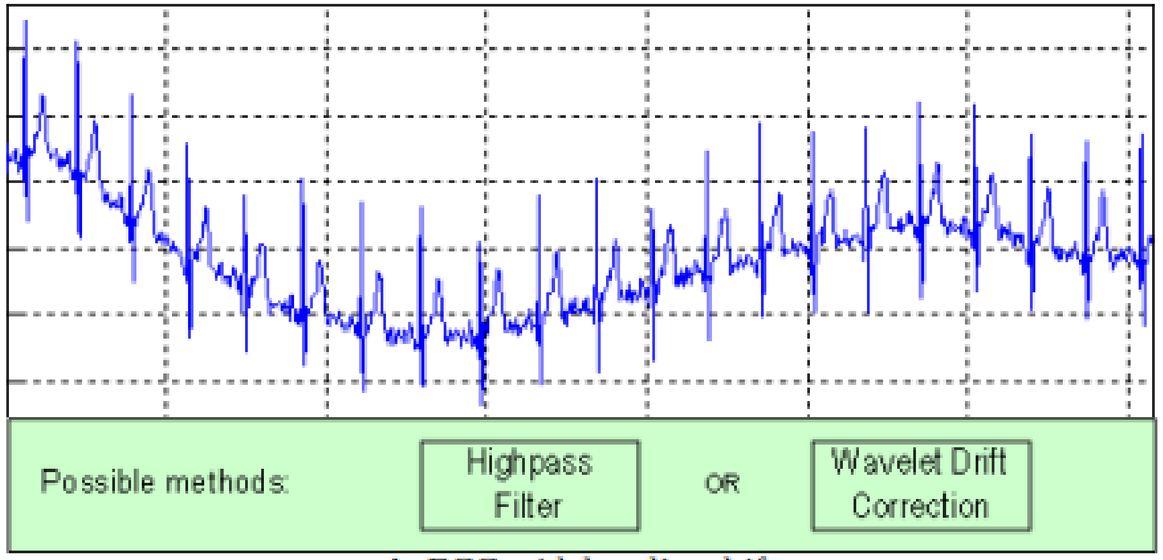
1 **Data Acquisition** : Single-lead ECG (Lead I: Left Arm – Right Arm), sample 500Hz, 16bit signal

2 **Data Preprocessing (Cleaning)** : Drift Correction for baseline wander (wavelet decomposition or high pass), Mains hum removal (Adaptive Bandstop/notch filter), HIgh frequency cleanup (low pass butterworth 50-80Hz)

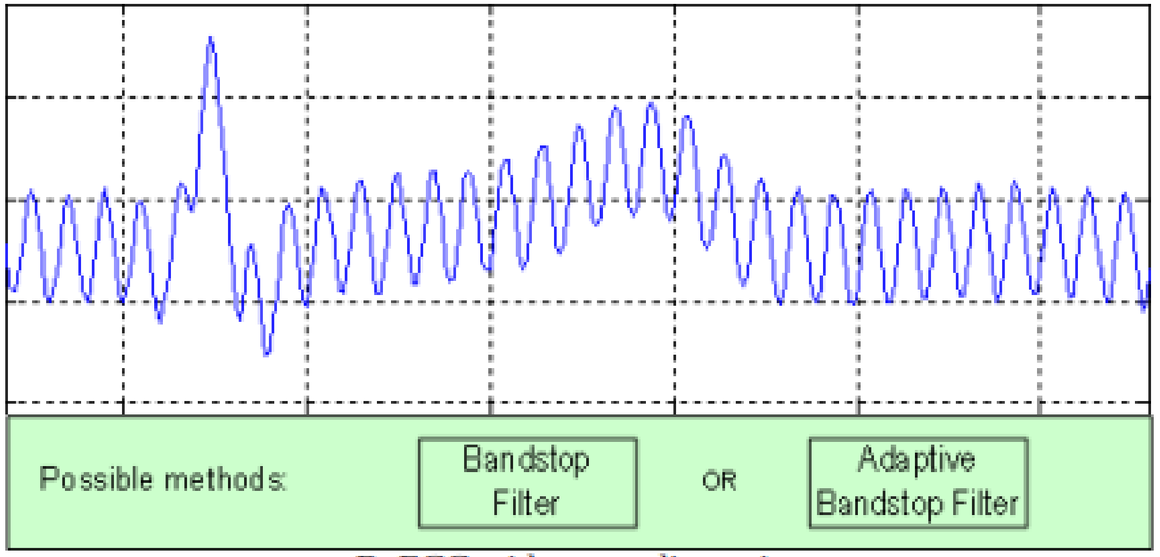
3 **Initial Feature Space Formation** : "Segmentation" phase where the continuous signal is chopped into individual heartbeats (cardiac cycle), Fragment Selection (P-QRS-T fragment out of QRS, P-QRS, QRS-T, and P-QRS-T windows provided the highest accuracy because the P-wave contains significant individual data), Window Size (window length of 0.5 seconds (250 samples) for every beat) aligned with R peak reference (80 samples before the peak and 170 samples after it), extracted 10 beats per record to choose a mean beat.

4 **Feature Space Reduction (PCA)** : Raw P-QRS-T window of 250 data points compressed into 30 pricipal components (vectors) using normalisation and dimensionality reduction for nullifying heart rate variations.

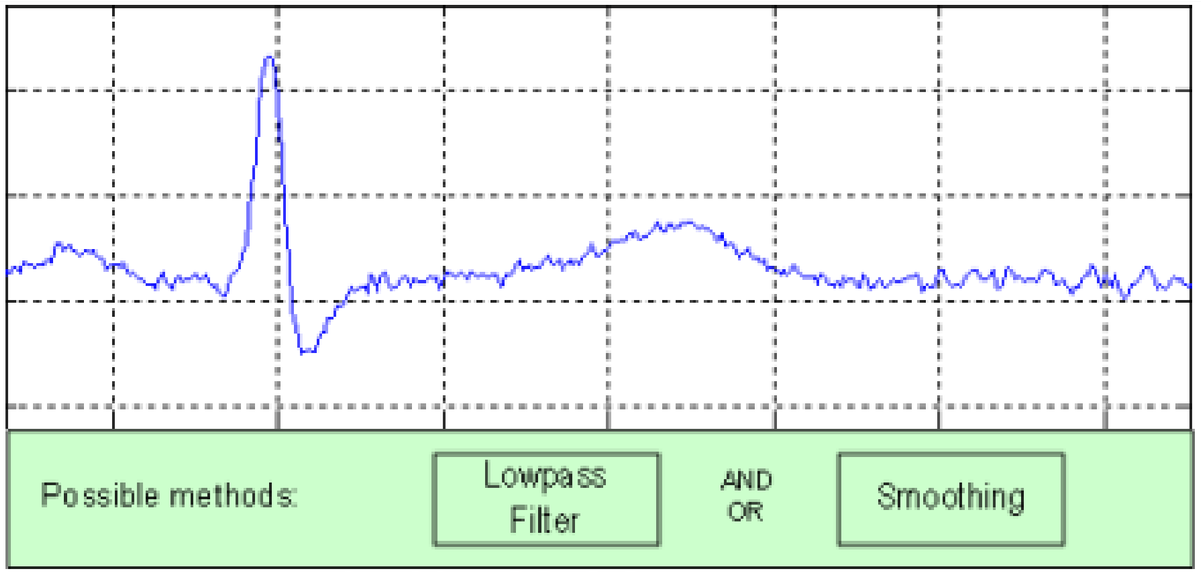
5 **Classification & Identification** : This final stage compares the user's 30-point "signature" against the database using Nearest Mean, Weighted Nearest Mean, or Linear Discriminant Analysis (LDA) algorithms with majority matches for multiple beats (cardiac cycle).



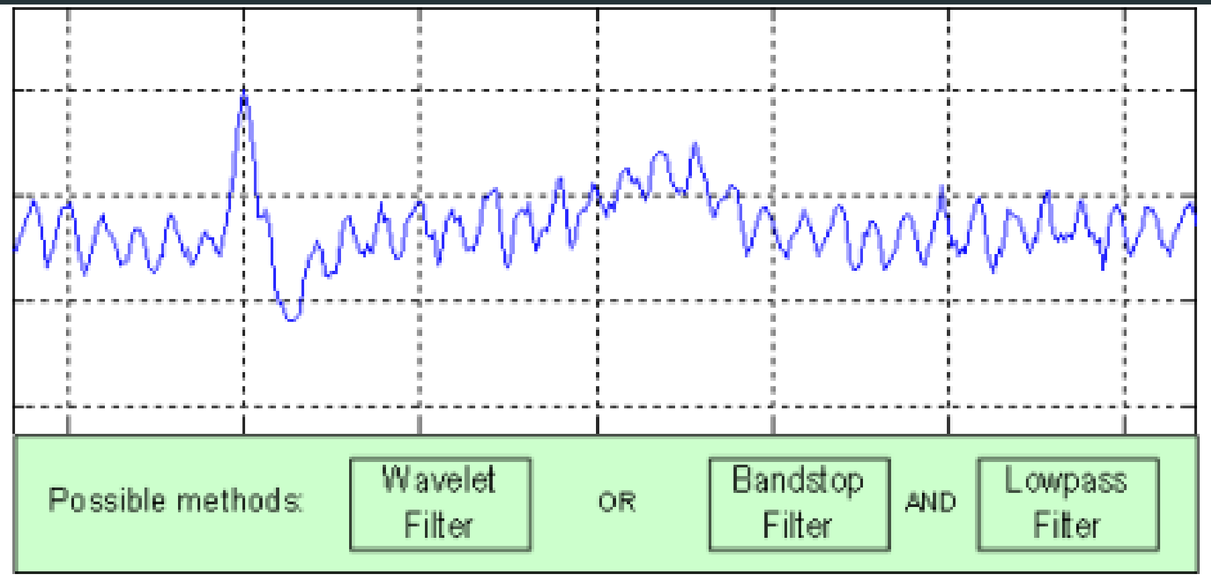
A. ECG with baseline drift



B. ECG with power-line noise



C. ECG with high-frequency noise



D. ECG with both power-line and high-frequency noise

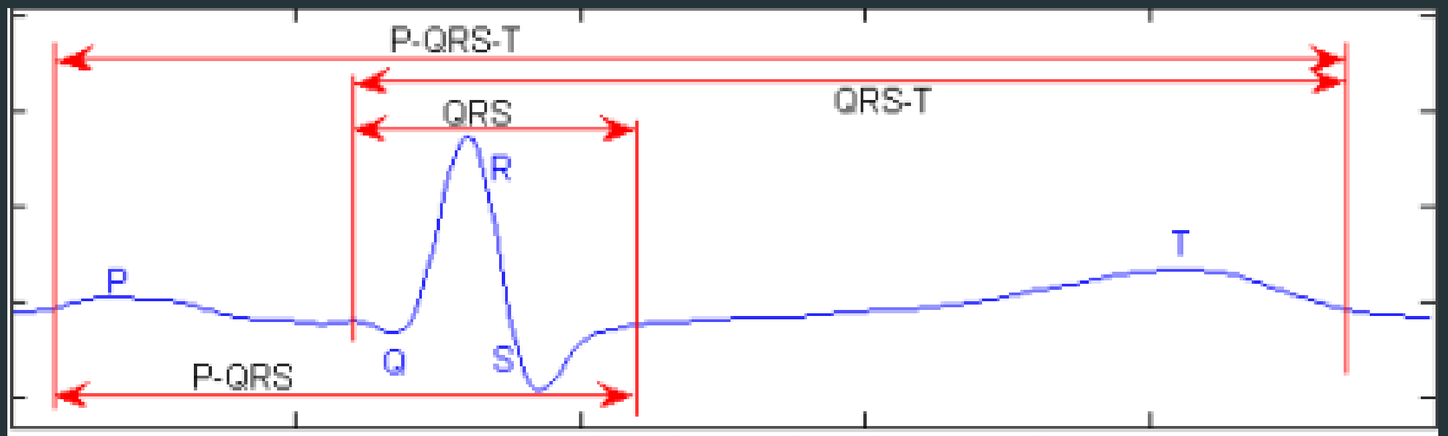


Figure 10. Variants of cardiac cycle informative fragments.

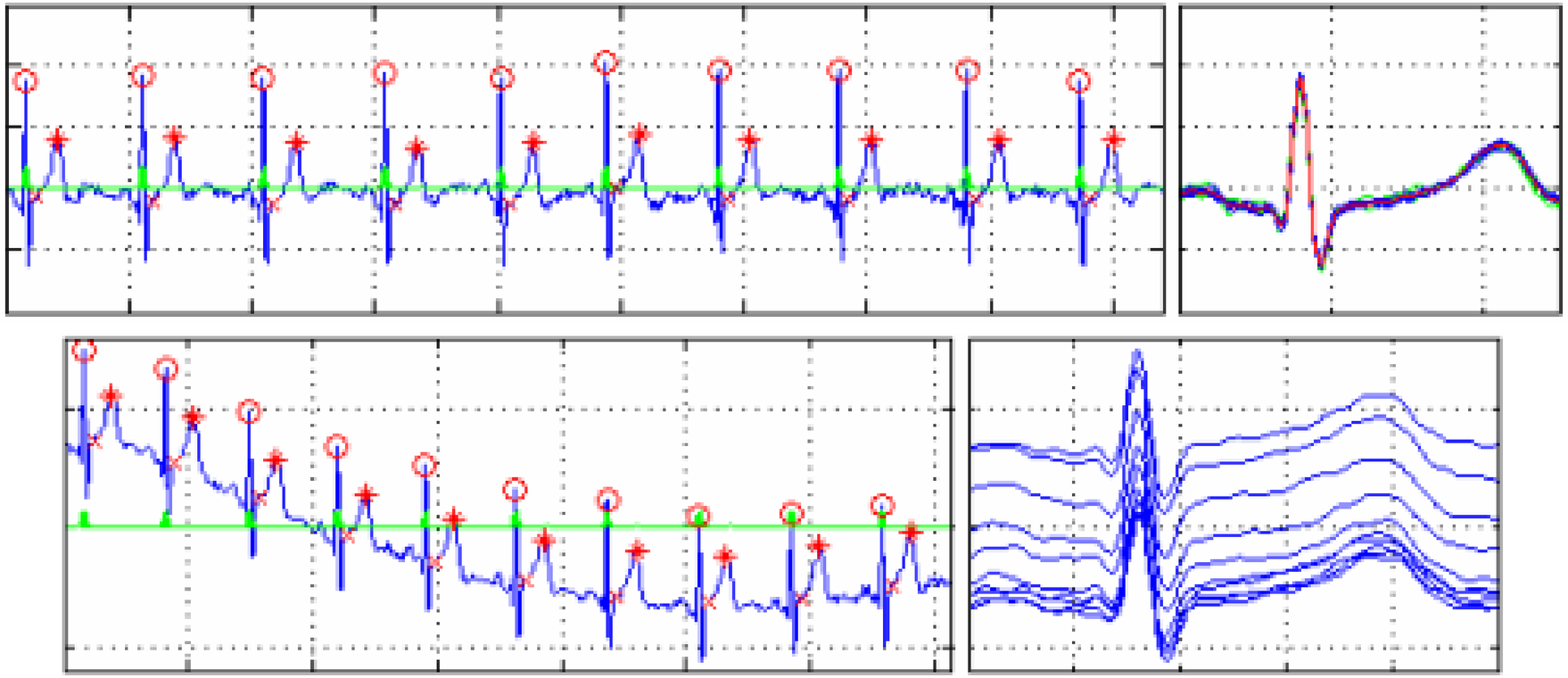


Figure 12. Examples of R peak detection and PQRST-fragment extraction.

Observations :

Study	Biel et al. [1]	Yi et al. [2]	Shen et al. [3]	[This study]
Number of individuals	20 (20-55 years)	9 (22-28 years)	20	90 (13-75 years)
ECG acquisition method	12 leads	Wireless, 30 minutes long	Lead I	Lead I, 20 seconds long
Number of ECG records	135 records, different days	18 records, different days	20 records	210 records, different days during 6 months
Number of ECG records for each individual	4-10 records	2 records	1 record	2-20 records
Training set	85 records	9 records of one day, 30 fragments for each record	20 heartbeats for each record	195 records, 6 of 10 heartbeats for each record
Test set	50 records	9 records of another day, all fragments for each record	1 heartbeat for each ECG (different part)	115 records, 6 of 10 heartbeats for each record
Informative features	Heartbeat waves amplitudes and intervals duration (N=30)	Coefficients of the wavelet decomposition of successive ECG fragments 10 seconds long	QRS complex and T wave amplitudes and intervals duration (N=7)	Samples of cardiac cycle fragment containing the QRS complex, P and T waves (N=250)
Reduction method	-	Principal Component Analysis	-	Principal Component Analysis or Wavelet Transform
Classification method	Soft Independent Modeling of Class Analogy (Multivariate Analysis)	Probabilistic neural network	Template matching (TM), Decision-based neural network (DBNN)	Linear Discriminant Analysis and Majority Vote Classifier
Identification results	98 %	95 %	TM: 95 %, DBNN: 80 %, Both: 100 %	96 %

Table 1. Summary of the key features of different studies.

Conclusion

Project Viability

- The "CardioKey" system demonstrates that the Electrocardiogram (ECG) is a viable, universal, and unique biometric trait for human identification.
- Research confirms that using a Single-Lead (Lead I) configuration can achieve identification accuracy rates as high as 96% under controlled conditions.

Advantages Over Traditional Biometrics

- ECG biometrics provides inherent "Liveness Detection," making it superior to fingerprint or face ID, which can be spoofed by static images or models.
- It offers a continuous method of authentication that is difficult to forge or steal.

Future Scope & Challenges

- The primary challenge remains Heart Rate Variability (HRV). As identified in recent studies, identification accuracy drops significantly (the "Rest-Ex" problem) if the user has just exercised.
- Future improvements will focus on advanced Normalization Algorithms (scaling the QT interval) to ensure the system remains robust even when the user's heart rate is elevated, also introducing more leads for accuracy.

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